

ACTUARIAL STATISTICS

Introduction

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Actuarial science

- **Actuarial science:** Discipline in which the risks covered by insurance companies are assessed through the application of statistical and mathematical techniques.
- **Risk:** Is the potential damage that can arise due to a current or future event. Colloquially, it is often used as a synonymous of probability. In a professional environment, the probability of occurrence of a negative event it is normally combined with how much damage this event would cause. In other words, risk is the possibility that a potential damage could occur.
- **Risk (business approach):** possibility that a future, incidental and economically unfavourable event could occur.

The insurance contract

- **Protection options in front of a risk:** prevention, saving or insurance.
 - **Prevention:** It does not eliminate the risk completely.
 - **Saving:** The risk can occur before accumulating the necessary capital.
 - **Insurance:** It covers the risk immediately and completely.
- **Insurance:** system by which a physical or legal person exposed to a risk gets, from a legal person, the commitment that the damage which may occur as a result of that risk would be compensated in exchange of a premium.
- **Insurance contract** (Law 50/1980, 8th October, on the Insurance Contract):
The insurance contract is the one by which the insurer assumes, by receiving a premium and in the case that the event (whose risk is covered) occurs, to indemnify, within the agreed limits, the damage caused to the insured or to satisfy a capital, income or other agreed benefits.

Types of insurance contracts

- **Types of insurance:** insurance can be sorted out depending on different criteria. Depending on the insured object they can be classified as personal insurance and property damage insurance.
- **Personal insurance:** the insured object is a person, the benefits payments depends on the existence, health or integrity of a person. Among them, there are the life insurance and health insurance.
- **Property damage insurance:** their main purpose is to compensate the policyholder for the suffered damage, which results from a claim that affects his/her property. Among them, there are: automobile and household insurance.

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- **Premium:** Price of the insurance. Is the amount of money that the policyholder pays so that, in exchange, the insurer will pay in case that the claim takes place.

$$\text{Premium} = \text{Pure_premium} + \text{Management_expenses} + \text{Tax} + \text{others}$$

- **Pure premium:** It comes from the concept of mathematical expectation or expected value of the real cost that the insurer assumes.
- **Actuarial statistics:** Branch of statistics that deals with the determination of the probabilities of the occurrence of the events which are covered by the policy.
- **Life actuarial statistics:** it focusses on life insurance contracts
- **Non life actuarial statistics:** it focusses on the rest of policies, such as the automobile or household insurance.

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- **Non life insurance:** the covered risks can take place **more than once** during the coverage period (**multiple events**) and has a **variable severity** (cost).
- **Life insurance:** the insured risk happens **only once** (**unique event**, the uncertainty is whether or not it happens during the coverage period) and the **severity is fixed** in the contract.
- For that reason, the statistical techniques used in life and non life insurance are utterly different. This is the reason for the existence of **life and non life insurance**.

Contents of the subject

- **Part 1: Life insurance**

- Introduction: the biometric model
- One life probabilities
- Two life probabilities
- Pricing life insurance contracts
- Mortality models

- **Part 2: Non life insurance**

- Modelization of the number of claims
- Modelization of the cost of claims
- Transformed random variables of interest in insurance
- Pricing models with GLM
- Other applications of GLM in insurance
- Risk measures